

2026 TURNAROUND MECHANICAL DECOKE

Cenovus Lima — Crude, Vacuum, HDS & Coker Furnaces

Job Report | USADebusk | Cenovus Lima Refinery — Lima, OH

PROJECT NO.	FACILITY	EXECUTION	PROJECT MANAGER
USA26046	Cenovus Lima Refinery	Sep 17 – 30, 2026	Jesse Utsey · Peter Campbell
WORK ORDERS	SCOPE	HEATERS	DURATION
Crude 82036534 · Vacuum 82036776 · Coker 82096135	TA Mechanical Decoke	4	14 days

4 HEATERS CLEANED	501.5 OPERATING HOURS	378 PIGS RUN	3 HEATERS SMART-PIGGED
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Project Information

Customer Details

FACILITY	Cenovus Lima Refinery
ADDRESS	1150 S Metcalf St, Lima, OH 45804
WORK ORDERS	Crude 82036534 · Vacuum 82036776 · Coker 82096135
CONTACT	Alfredo Trevino (Crude, Vacuum, HDS) · Kaleb Dodson (Coker)

Project Details

SCOPE	2026 turnaround mechanical decoke of four fired heaters, with filtration and smart-pig support
EXECUTION	September 17 – 30, 2026 (14 days)
HEATERS	PR-170028 Crude · PR-170029 Vacuum Charge · PR-175169 HDS · PR-170002 Coker
EQUIPMENT	(2) Trimax Pumpers (Trimax 5 & 6) · (2) Support Units · (2) Filter Presses · Crew Trucks

Crew Details

PROJECT MANAGER	Jesse Utsey (Vacuum) · Peter Campbell (Crude, HDS, Coker)
SHIFT LEAD	Day — Jesse Utsey, Peter Campbell; Night — Sam Mixon, James McDaniel
DAYSHIFT	Danilo Ramirez, Blake Harman, Blake Baugh, Rodney Lynch, Brody Lebleau, David Charles, Adrian Garza, Marshall Douglas, Travis Trenholm
NIGHTSHIFT	Joel Ruiz, Tyler Garza, Justin Kelly, Mikee Green, Jeremiah Harman

Project Duration

SCOPE	UNIT	RIG	PIG	SMART PIG	SUBTOTAL
PR-170028 Crude Furnace	Trimax 5	35.75	118	38.25	192
PR-170029 Vacuum Charge Furnace	Trimax 6	58.5	63	74	195.5
PR-175169 HDS Furnace	Trimax 5	16	20.5	4.5	41
PR-170002 Coker Furnace	Trimax 5	22	51	0	73
TOTAL		132.25	252.5	116.75	501.5

Stand-By Summary

Stand-by is on-site time during which the pumping units were not actively pigging. Cause is recorded from the daily service receipts.

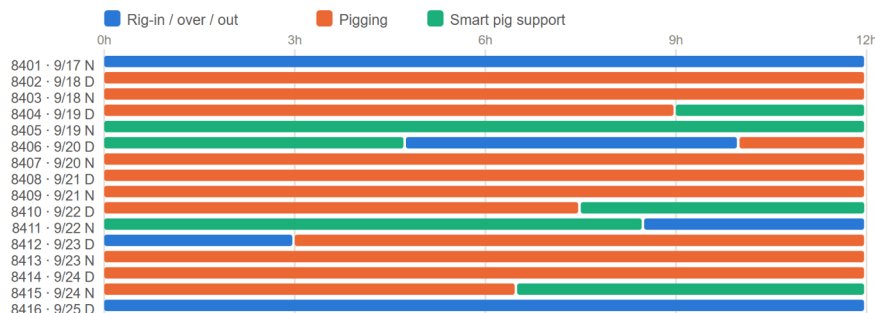
DATES	HOURS	CAUSE
Sep 19 – 21 (7077–7081)	49	Awaiting installation of the 12" radiant-outlet launchers (Vacuum)
Sep 26 (7091)	3	Awaiting relocation of the 12" launchers to passes 5, 7, 8 (Vacuum)
Sep 25 (8417)	1	Site safety review of hose routing near the catalyst job (HDS)
Sep 27 (8420)	1	Awaiting a 1600 CFM compressor to dewater (HDS)
TOTAL	54	

Combined project total: 501.5 operating hours plus 54 stand-by hours — 555.5 pumping-unit hours.

PR-170028 — Crude Furnace

Number of Passes	8, pigged individually
Total Footage	23,856 ft (2,982 ft per pass)
Convection Tube ID	5.761" (6" Sch 80S)
Radiant Tube ID	7.625" (8" Sch 80S)
Metallurgy	A312 TP317/317L
Return Bends	180° short-radius
Smart Pigging	Yes — Intero

Decoking Analysis: Fouling was concentrated in the upper 8" radiant tubes, tubes 1 through 5, with lighter fouling through the balance of the coil. Average pigging time was approximately 39 hours per pass.



PR-170028 - task hours by shift, from the daily service receipts.

Pigs Used

SIZE — QTY	SIZE — QTY	SIZE — QTY	SIZE — QTY
6" — 30	6.25" — 50	8" — 40	8.5" — 12
6.125" — 12	7.75" — 3	8.25" — 54	
TOTAL — 201			

Flow Tests

Pass 1

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	310	270	40
700	285	255	30
650	250	225	25

Pass 2

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	315	280	35
700	295	270	25
650	270	255	15

Pass 3

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	350	305	45
650	275	220	55
550	220	165	55

Pass 4

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	324	295	29
650	248	230	18
550	177	165	12

Pass 5

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	270	245	25
650	215	190	25
550	155	125	30

Pass 6

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	300	285	15
700	280	255	25
600	240	205	35

Pass 7

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	290	265	25
700	265	240	25
600	230	205	25

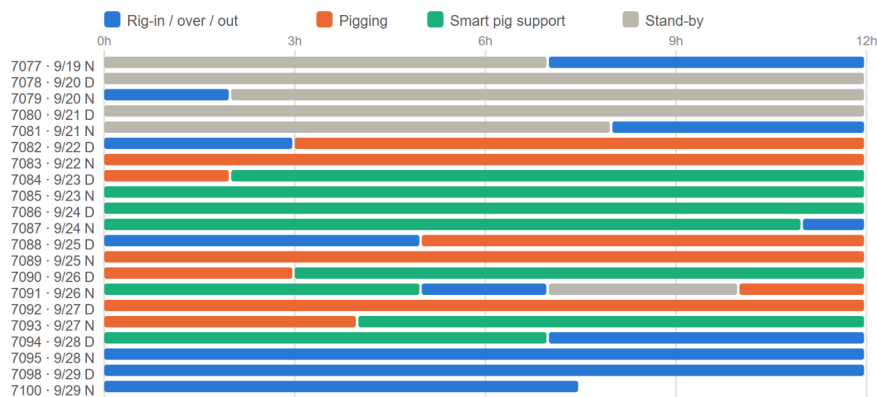
Pass 8

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	250	221	29
700	220	190	30
600	200	165	35

PR-170029 — Vacuum Charge Furnace

Number of Passes	8, pigged individually
Total Footage	7,054 ft (882 ft per pass)
Convection Tube ID	4.000"
Radiant Tube ID	3.826" · 5.761" · 7.813" · 9.750" · 11.750" (4" to 12")
Metallurgy	A312 TP317/317L
Return Bends	180° seamless
Smart Pigging	Yes — Intero

Decoking Analysis: Passes 4 and 5 carried the heaviest fouling, pass 5 the most. In pass 5, fouling was heavy in the 12" section, moderate in the 10" and 4" sections, and light in the 6" and 8" sections. In pass 4, it was moderate in the 12", light to moderate in the 10", and light in the 8", 6" and 4" sections. The remaining six passes showed light fouling throughout. The external 6" convection inlet piping, between the control valve and the 4" convection inlet tube, was heavily fouled on all eight passes, enough to keep 4" pigs from entering the convection section, and was cleaned first on each pass. Average pigging time was approximately 21 hours per pass.



PR-170029 - task hours by shift, from the daily service receipts.

Pigs Used

SIZE — QTY	SIZE — QTY	SIZE — QTY	SIZE — QTY
3.75" — 4	4.125" — 15	6.25" — 6	10" — 20
3.8" — 1	4.25" — 4	8" — 7	12" — 6
4" — 22	6" — 10	8.25" — 2	
TOTAL — 97			

Flow Tests

Pass 1

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	295	220	75
650	260	185	75
600	225	165	60

Pass 2

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	275	225	50
650	235	195	40
600	205	170	35

Pass 3

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	332	282	50
650	290	250	40
600	255	220	35

Pass 4

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	280	210	70
650	235	188	47
600	210	160	50

Pass 5

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	270	220	50
650	235	190	45
600	202	160	42

Pass 6

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	285	225	60
650	235	195	40
600	210	165	45

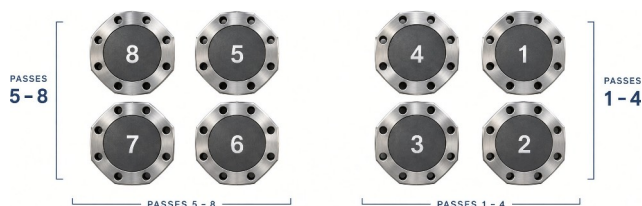
Pass 7

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	260	220	40
650	225	185	40
600	195	160	35

Pass 8

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	320	280	40
650	280	250	30
600	240	220	20

Images



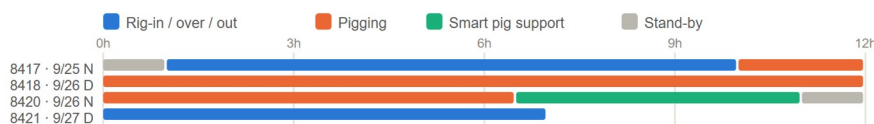
Radiant outlet labels and orientation, viewed facing the outlets. Each outlet connects to the convection inlet of the same pass number.

Top: coke removed from the 12" tube, pass 5. Bottom: deposits removed from the external 6" convection inlet piping, collected from all eight passes.

PR-175169 — HDS Furnace

Number of Passes	4, looped into 2 circuits (passes 1 + 2, passes 3 + 4)
Total Footage	3,600 ft (900 ft per pass)
Radiant Tube ID	5.885"
Metallurgy	A335 P5 (radiant)
Smart Pigging	Yes

Decoking Analysis: Light fouling was found throughout the convection and radiant sections. Average pigging time was approximately 20.5 hours per circuit.



PR-175169 - task hours by shift, from the daily service receipts.

Pigs Used

SIZE — QTY	SIZE — QTY
6" — 4	6.25" — 12
TOTAL — 16	

Flow Tests

Passes 1 + 2 (looped)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	320	267	53
750	290	238	52
700	250	215	35

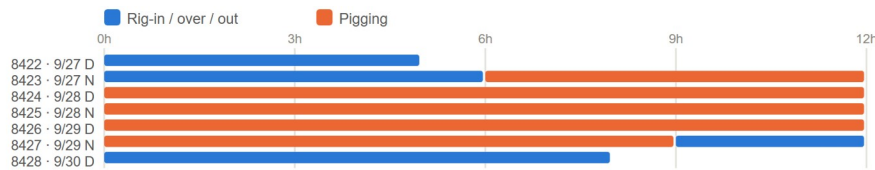
Passes 3 + 4 (looped)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
650	230	180	50
600	210	170	40
550	175	140	35

PR-170002 — Coker Furnace

Number of Passes	4, pigged as 3 circuits: convection (4 coils looped as one), radiant A + B, radiant C + D
Total Footage	15,904 ft (3,976 ft per pass)
Tube ID	3.398" (4" OD tube); radiant outlet section 3.624" (4" Sch 120)
Metallurgy	9Cr-1Mo
Return Bends	Cast 180° U-bends

Decoking Analysis: The convection section showed light fouling and the radiant section moderate fouling. The crossovers were removed, the convection was pigged as a single circuit, and the radiant passes were looped A + B and C + D with temporary 180° returns at the radiant inlet flanges. Average pigging time was approximately 51 hours per circuit.



PR-170002 - task hours by shift, from the daily service receipts.

Pigs Used

SIZE — QTY	SIZE — QTY	SIZE — QTY	SIZE — QTY	SIZE — QTY
3" — 3	3.25" — 6	3.325" — 3	3.625" — 21	3.875" — 4
3.125" — 3	3.3" — 2	3.5" — 6	3.75" — 14	4" — 2
TOTAL — 64				

Flow Tests

All convection

GPM	BEFORE PSI	AFTER PSI	Δ PSI
350	435	375	60
300	330	290	40
250	220	180	40

Passes C + D

GPM	BEFORE PSI	AFTER PSI	Δ PSI
350	419	287	132
300	320	219	101
250	225	159	66

Project Summary

USADebusk performed the mechanical decoke of four fired heaters for Cenovus at the Lima Refinery during the 2026 turnaround, September 17 – 30: the Crude Furnace (PR-170028), Vacuum Charge Furnace (PR-170029), HDS Furnace (PR-175169) and Coker Furnace (PR-170002). Two Trimax pumping units ran independent schedules — Trimax 6 on the Vacuum heater, and Trimax 5 on the Crude, HDS and Coker in turn. The Crude, Vacuum and HDS heaters were smart-pigged after cleaning. Fifty-two of the project's 54 stand-by hours were on the Vacuum heater, waiting on the 12" radiant-outlet launchers to be installed and later relocated.

Project Close

All four heaters were cleaned and returned to Cenovus, and all equipment was demobilized from the facility on September 30, 2026.

USADebusk thanks the Cenovus Lima team for their coordination and support throughout the turnaround, and we look forward to continuing to serve your decoking and pigging needs.

Jesse Utsey | Project Manager, USADebusk

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